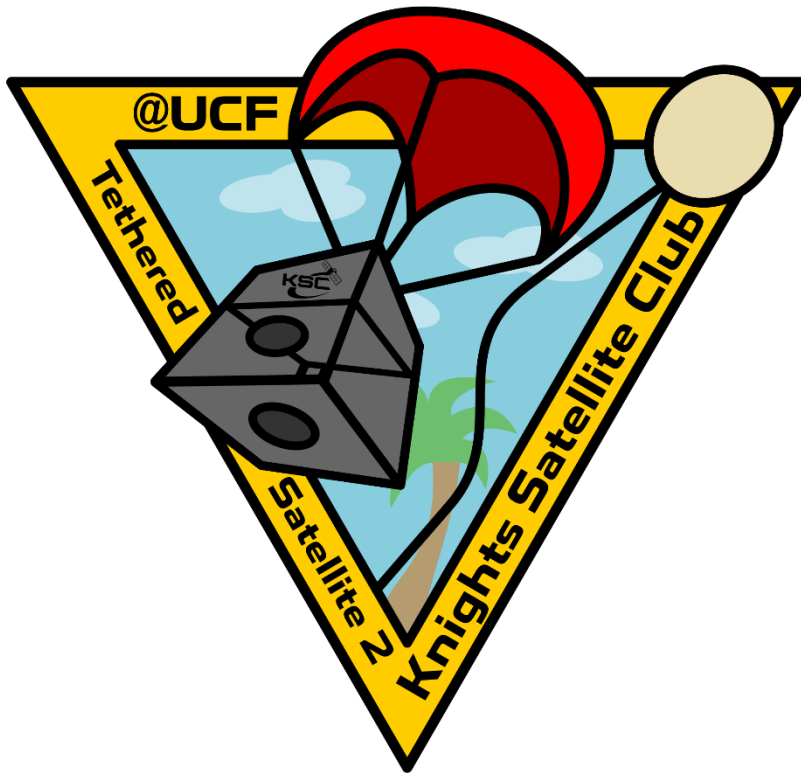




Flight Readiness Review (FRR)

Tethered Satellite 2 Mission A

Knight's Satellite Club



Introduction and Flight Readiness Assessment

Purpose and Scope

The Flight Readiness Review (FRR) for Tethered Satellite 2 Mission A (T-Sat 2A) documents the final preparation, launch operations, flight performance, recovery, and postflight assessment of the mission developed by the Knight's Satellite Club (KSC) at the University of Central Florida. Building upon the design and engineering documentation presented in the Critical Design Review (CDR), this document focuses on the operational activities performed during the final week leading up to launch, the readiness of the integrated vehicle and ground station, and the results observed during the mission. The purpose of this review is to preserve the engineering decisions, operational experiences, and lessons learned from the completed flight so that future KSC teams can improve the reliability of subsequent Tethered-Satellite missions.

The T-Sat 2A mission was developed to demonstrate an integrated 2U CubeSat-style system incorporating imaging, altitude measurement, wireless communication, onboard data storage, and a remotely actuated separation mechanism. Although the system successfully collected flight data and demonstrated operation of its integrated electronics, the mission did not achieve the intended parachute-assisted descent. This FRR documents both the successful and unsuccessful aspects of the flight without treating the mission outcome as a substitute for a complete technical assessment of each subsystem.

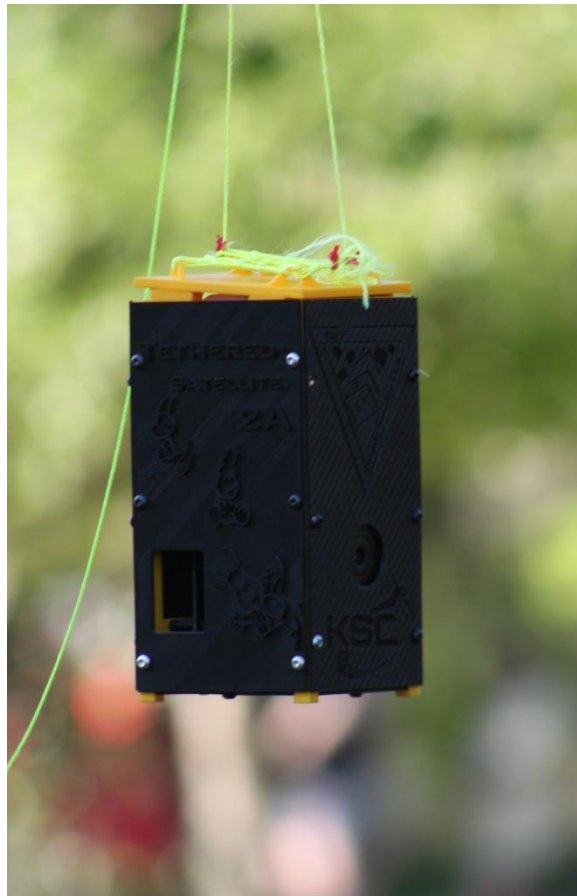


Figure 1: Launch Ready T-Sat 2A

Final Week of Launch Preparation

Preflight Testing and Initial Findings (March 24, 2026)

The final week of launch preparation began on March 24 with testing of the flight electronics and deployment system. The battery was subjected to a continuous 20-minute test that included multiple servo activations, during which no data corruption was observed. This test provided an initial indication that the electrical system could maintain operation while the servo was being activated. However, the servo blade did not successfully cut the fishing wire during the mechanical release test. The team identified the need to improve the cutting mechanism, including the possibility of adding an additional support hole and using a newly sharpened blade to improve the cutting performance.

The unsuccessful cutting test identified a critical risk to the separation system. Since the servo's operation alone did not guarantee that the fishing wire would be cut, the team needed to address both the physical support of the actuator and the positioning of the wire relative to the blade before proceeding with launch. The results of this test became the basis for additional mechanical modifications during the following days.

Power-System Troubleshooting (March 25, 2026)

On March 25, the team identified an issue associated with low battery output during testing. The reduced battery output affected the team's confidence in the available power for continued testing and deployment operation. In response, the team adjusted its schedule to allow the batteries to be charged earlier for the following day's testing. This event reinforced the importance of maintaining adequate battery charge and verifying power availability before conducting tests involving the servo and integrated electronics.

The power-related issue was addressed as part of the continuing preparation process, although the available notes do not establish a complete electrical root cause or provide measured voltage values from the event. Consequently, the issue is documented as an observed testing concern rather than a conclusively identified battery failure.



Figure 2: T-Sat 2A Final Weight

Mechanical Reinforcement and Integrated System Testing (March 26)

On March 26, the team continued addressing the release mechanism's mechanical limitations. Epoxy was purchased to secure the servo in its mounting position because the servo required additional physical support and pressure to achieve a successful cut. Additional bolt holes were incorporated into the structure to provide improved mechanical support for the servo and its mounting interface. These changes were intended to reduce actuator movement during operation and improve the alignment between the cutting blade and fishing wire.

Following the mechanical modifications, the team performed a dry test of the fully assembled PCB system within the CubeSat housing. This test allowed the team to examine the integration of the electronics and mechanical structure while also evaluating the radio communication distance through the housing. Testing the integrated configuration was important because the final flight system required the mechanical housing, onboard electronics, and communication system to operate together rather than only as individually tested components.

Final Integration and Launch-Eve Preparation (March 27, 2026)

The final day of testing involved troubleshooting the radio communication and preparing the CubeSat for launch. The team identified a power-related concern associated with the radio signal and observed instances where the transmitted signal was not received on the first attempt. To address the command-handling behavior, the software was modified so that pressing the ground-station button would clear incoming data and messages before focusing on the RFM LoRa signal associated with the release command. This modification was intended to reduce interference from other incoming messages and prioritize the deployment command.

The team also finalized the rope connections between the top panel and the balloon assembly, including the knots intended for launch. During the final mechanical preparation, the servo panel floor fractured. Because the component was required for the launch configuration, a replacement was scheduled to be printed by 7:00 a.m. on the following morning. This introduced an additional time constraint immediately before launch and required the team to complete a final replacement, integration, and verification process within a limited period.

To address movement observed during testing, the team used 3D-printed material from spare T-Sat housing parts to create a barrier and improve the fit between the top panel and the housing. A soldering pen was used to melt and shape pieces of the spare printed material, allowing the team to reduce unwanted movement between the components. The resulting fit was described as snug, with minimal movement during separation testing. The purpose of the modification was to limit unintended displacement of the top panel and reduce the likelihood that the fishing line would contact the cutting blade prematurely.

The team charged four batteries for launch, with two batteries intended as spares. The preparation of spare batteries provided an alternative power source if a battery-related issue arose during the final launch sequence. The combined activities of March 27 represented the final mechanical, electrical, software, and logistical preparation before the vehicle was transported and assembled for launch.

Launch Day Operations — March 28, 2026

Team Coordination and Launch Preparation

Launch-day operations involved coordination between the T-Sat A team, the T-Sat B team, and the Balloon Launch Regulators. The launch team arrived with the objective of completing the final assembly, verifying the electrical system, connecting the CubeSat to the balloon assembly, and preparing the ground station for operation. The T-Sat A team was ready to begin launch preparation at approximately 9:00 a.m., following an original meeting time of 8:00 a.m. The additional preparation period was used to complete the final assembly and launch configuration.

Upon arrival, epoxy was applied to the servo assembly to secure its position. The team allowed approximately one hour of curing time before proceeding with the remaining preparations. The batteries were checked for full voltage, and the CubeSat panels were closed while leaving one panel temporarily open to provide access to the electronics and servo-related connection points. This configuration allowed the team to connect the battery and complete the final fishing-line attachment without unnecessarily leaving the electronics powered for an extended period before launch.

The fishing line was initially attached loosely to the housing so that the connection could be adjusted when the balloon assembly was connected. Once the launch preparations were completed, the line was secured to the intended attachment points. The top panel was configured to remain attached to the balloon following separation, while the main CubeSat housing was intended to separate and descend with the parachute. The team completed the final knotting process and verified the physical arrangement of the tether and panel connections.

Final Electrical and Ground Station Checks

Before the final panel was closed, the team connected the ground station to a laptop and transmitted a test signal without initiating a physical deployment. This test was intended to verify that the ground-station software was running the expected code and that the communication system was prepared for launch operations. The ground station was used to monitor altitude information and issue the remote release command through the RFM LoRa communication link.

The team maintained access to the electronics through a panel that remained open during the final stages of preparation. This access allowed the battery connections and fishing-line attachment to be completed shortly before launch. The final panel was closed after the team heard the camera shutters and the ground-station personnel confirmed the altitude readings. The sequence was intended to reduce the time between connecting the batteries, sealing the housing, and releasing the balloon.

The final preparation process required balancing the need for access to the hardware against the need to minimize the time the system remained powered before launch. The use of a temporarily open panel provided the team with access to the internal components, but the limited time available for final integration also increased the importance of completing the last assembly steps without introducing additional mechanical or electrical errors.

Flight and Deployment Operations

Balloon Release and Initial Ascent

Once the launch team determined that the system was ready, the balloon was released and the flight sequence began. Following release, the CubeSat experienced some bouncing and movement between the top panel and the main housing. This behavior caused concern among the team because the separation mechanism depended on the stability and alignment of the physical connections. Despite the observed movement, the launch continued and the CubeSat remained connected to the balloon assembly.

During the ascent, a drill attachment intended to unreel the rope spool at a continuous speed failed. The failure caused the rope to become tangled around itself and resulted in the CubeSat remaining at an altitude of approximately 20 ft. The initial spool-based deployment method therefore did not maintain the intended rope payout. After the tangling problem was addressed, the balloon launch team manually reeled out the remaining rope. This allowed the ascent to continue at a more consistent and slower rate.

The use of a manual rope-reeling method was an operational adjustment made in response to the failure of the original spool mechanism. Although the revised method allowed the mission to continue, it changed the ascent behavior and introduced an additional manual task for the launch team. The event demonstrated that the launch system depended not only on the CubeSat's internal hardware but also on the proper operation of the external balloon and tether equipment.

Altitude Monitoring and Separation Command

After the rope issue was resolved, the CubeSat continued ascending while tumbling beneath the balloon assembly. The flight continued for approximately 20 minutes and reached an altitude of around 450 ft based on the altitude readings observed by the team. The altitude sensor's measurements were reported in meters, requiring the launch team to use the corresponding altitude information when determining when to initiate the separation command.

Based on earlier altitude tests, the team had expected the altitude readings to respond within approximately 20–30 seconds during descent or changing altitude conditions. However, as the balloon assembly moved farther away, the altitude readings took approximately one minute or longer to update. The observed delay affected the timing of the release decision and required the team to wait for confirmation of the intended altitude before sending the separation command.

Once the altitude was confirmed at approximately 450 ft, the operator pressed the ground-station release button. The ground station transmitted the separation command through the RFM LoRa communication system. Although an acknowledgement message was received, the initial servo actuation did not successfully cut the fishing wire. The system then proceeded through the backup command-handling behavior, which prioritized the servo cutting function after receiving another type of message through the radio system. This resulted in the CubeSat eventually becoming disconnected from the weather balloon assembly.

The successful disconnection demonstrated that the communication system and servo control software were able to participate in the release sequence. However, the initial failure of

the blade to cut the fishing line showed that receiving a deployment command and completing the intended physical separation were separate outcomes. The mission therefore exposed a distinction between software-level acknowledgement and verified mechanical deployment.

Parachute Deployment and Descent

Following the eventual separation of the CubeSat from the balloon, the parachute deployment sequence occurred over approximately eight seconds. The parachute began attempting to exit through the open top panel, but it remained inside the housing for approximately six seconds. During the final two seconds of the sequence, the parachute was observed to be partially extended outside the housing. The parachute did not achieve sufficient deployment to provide the intended lift or controlled descent for the CubeSat.

The incomplete parachute deployment prevented the descent system from performing its primary function. The observed sequence indicated that the parachute was able to begin moving toward the exit, but it did not fully clear the housing in the available time. The available observations do not establish the exact point at which the parachute became obstructed or the complete mechanism responsible for preventing deployment. The postflight inspection provided additional evidence regarding the physical interaction between the parachute and the printed structure.



Figure 3: The Remains of T-Sat 2A

Recovery and Postflight Inspection

Physical Recovery and Condition of the Vehicle

After the CubeSat descended, the launch team recovered the damaged hardware and collected the separated components from the UCF Arboretum area. The recovery process included gathering the broken housing pieces and preserving them for further inspection. The team retained the damaged parts to help understand the events that occurred during the descent and impact, as well as to identify mechanical features that could be improved in a future version.

The recovered housing showed substantial structural damage. One section of the 2U wall remained attached through its screw connection to the approximately 1.5U top panel and a portion of the bottom-panel connection. Two panels were cracked, while the opposite wall was split into two pieces. Despite the structural damage, the cameras were not damaged, and the two PCBs appeared to remain physically intact. The onboard modules also appeared undamaged based on the team's initial inspection. The SD card remained present and usable, while some components had become disconnected from their standoffs and crimp connections. The camera crimp connections remained attached to the PCB boards.

The panel identified as Project A.L.E.X. remained undamaged during the recovery. This component was retained along with the other recovered parts for inspection and documentation. The condition of the vehicle demonstrated that the printed housing experienced significant damage during the fall, while several internal electronic components and data-storage elements remained sufficiently intact to allow postflight data retrieval.



Postflight Data Recovery and Battery Assessment

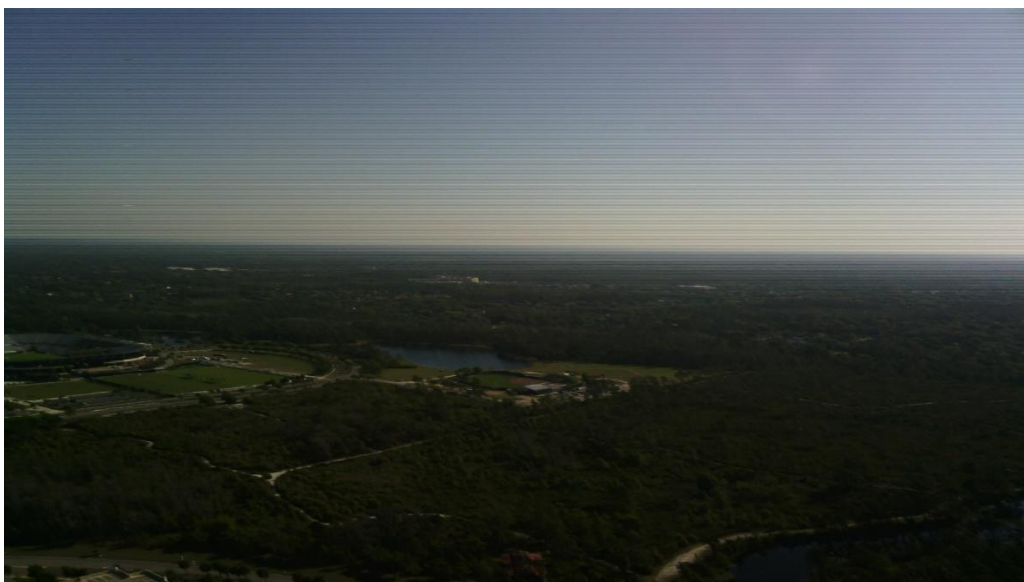
Following recovery, the team inspected the SD card and found that it remained in good condition. The stored files included images captured during the flight and altitude data recorded during the mission. The image data provided a record of the flight, while the altitude information remained available for review until what the team believed to be near the peak altitude of the flight. These data provided an opportunity to reconstruct portions of the mission and compare the observed flight behavior with the intended operating sequence.

The continued availability of the onboard data was particularly important because the radio communication system did not provide a continuous record of every event. The SD card preserved information that could be reviewed independently of the ground station's received telemetry. The data therefore supported the postflight assessment of the camera and altitude-recording systems, although a complete quantitative analysis of the recovered data would be required to establish the precise duration, maximum altitude, and final measurement quality.

The recovered batteries retained approximately 3.7 V during the postflight inspection and appeared to be in suitable physical condition for further assessment. However, the batteries were not automatically considered safe for reuse solely on the basis of their measured voltage or physical appearance. Additional testing would be necessary before returning them to service, including an evaluation of their condition following the flight and any mechanical damage or electrical stress experienced during operation.



Figure 4: Images Captured by T-Sat 2A



Mission Results and Assessment

Mission Objectives and Observed Results

T-Sat 2A successfully demonstrated the integration of a 2U CubeSat-style housing, onboard electronics, imaging systems, altitude sensing, local data storage, wireless communication, and a remotely controlled separation mechanism. The mission reached an observed altitude of approximately 450 ft, and the CubeSat eventually separated from the balloon assembly after the initial release attempt failed to cut the fishing wire. The onboard cameras and altitude-recording system preserved data that could be retrieved after the flight, demonstrating that portions of the integrated system continued operating sufficiently to record and store information during the mission.

The intended parachute-assisted descent was not achieved. The parachute remained inside the housing for most of the observed deployment sequence and only partially exited during the final portion of the event. As a result, the CubeSat did not experience the planned controlled descent, and the housing sustained substantial structural damage during the fall. The separation and descent sequence therefore produced mixed results: the system eventually achieved physical disconnection from the balloon, but the intended deployment and recovery behavior was not completed.

The mission provided evidence that several major functions of the integrated system could operate in a real flight environment. However, successful operation of individual components did not ensure that the complete mechanical release and parachute deployment sequence would perform as intended. The mission results highlight the need to assess the entire operational chain, from the release command and cutting mechanism through the physical separation, parachute exit, and descent.

Suspected Cause of Parachute Deployment Failure

Based on the postflight inspection, the team identified the newly printed servo panel as a suspected contributing factor in the unsuccessful parachute deployment. During the final launch preparation, the support infill beneath the arch of the printed component was removed. This infill had been intended to allow the fishing wire to pass directly through the structure. The removal of the support material changed the physical configuration of the area surrounding the fishing line and the parachute exit path.

The postflight inspection identified a tear in the parachute that appeared to be associated with the parachute's attempted exit from the housing. A portion of the 3D-printed support infill was found to remain attached to the parachute. This observation provided evidence that the remaining printed material may have interfered with the parachute during deployment. The team believes that the retained support material contributed to the parachute becoming caught as it attempted to exit the housing. This explanation is consistent with the observed tear and the presence of printed material on the parachute, but it should be treated as a suspected cause rather than a conclusively demonstrated root cause without further testing.

The issue was not identified during the final knotting and attachment process because the affected material was located on the opposite side of the servo relative to the fishing-line knot. The team had positioned the knot to avoid interference with the cutting blade, and this

positioning limited visibility of the material that remained in the parachute's potential exit path. The event demonstrates the importance of inspecting the entire deployment path, including surfaces that are not directly involved in the cutting mechanism, before closing the final flight configuration.

Engineering Assessment and Lessons Learned

The T-Sat 2A mission provided the team with experience in developing and operating an integrated system within a limited development timeline. The project brought together mechanical design, electrical integration, software development, manufacturing, radio communication, and flight operations. The successful recovery of flight data and the continued condition of several electronic components demonstrated that the mission produced useful engineering results even though the planned parachute descent was not achieved.

One of the most significant lessons from the mission was the importance of maintaining sufficient time for final configuration verification. The servo panel was replaced on the morning of launch after the original panel floor fractured. The replacement and subsequent modifications were completed under a highly compressed schedule. Although the team performed final preparations and testing, the available time limited the opportunity to inspect the replacement component and its surrounding geometry in the same depth as a more deliberate integration process. The suspected interference between the remaining printed support material and the parachute illustrates how a small manufacturing or assembly detail can affect the performance of the complete deployment system.

The mission also reinforced the need to distinguish between a successful electronic command sequence and successful mechanical execution. The system received a deployment command and eventually separated the CubeSat from the balloon, but the initial servo activation did not cut the fishing wire. In addition, the parachute deployment was not successful despite the physical separation. Future missions should therefore include verification procedures that evaluate the entire release and deployment path, rather than relying solely on software acknowledgements or servo movement as evidence of successful deployment.

The failure of the rope-reeling attachment during ascent provided another operational lesson. The intended continuous-speed spool mechanism did not perform as expected, resulting in a rope tangle and a temporary halt in ascent. The team was able to resolve the issue through manual intervention, but the event showed the value of verifying external launch equipment and establishing contingency procedures before the flight. The operational response allowed the mission to continue, although it changed the intended ascent behavior and required additional coordination between the launch team and the balloon operators.

Final Mission Assessment and Recommendations

Overall Mission Assessment

T-Sat 2A completed a flight that demonstrated the integration and operation of multiple subsystems within a student-developed CubeSat platform. The mission reached an altitude of approximately 450 ft, collected image and altitude data, and eventually separated from the balloon assembly. The flight also exposed failures in the mechanical cutting system, the external rope-reeling equipment, and the parachute deployment sequence. The recovered hardware and data provided the team with information that can be used to improve the mechanical and operational reliability of future Tethered-Satellite missions.

The mission should be considered a completed engineering flight with both successful subsystem operation and an unsuccessful planned descent outcome. The flight results do not invalidate the development work performed by the team, but they do identify areas where the final integration and launch-readiness process must be strengthened. The recovered components and recorded data provide a foundation for future design changes and further verification testing.

Recommendations for Future T-Sat Missions

Future T-Sat teams should prioritize a complete final configuration inspection after any replacement, modification, or repair performed during the final days before launch. In particular, the entire parachute exit path should be checked for support material, sharp edges, protrusions, and other features that could interfere with deployment. This inspection should be performed with the parachute installed in its flight configuration rather than only evaluating the cutting mechanism independently.

The separation system should also be tested with the final flight hardware and attachment geometry. Testing should verify that the servo remains mechanically supported, that the fishing wire is properly aligned with the blade, and that the release mechanism can complete its intended physical action. Where possible, the team should conduct a repeatable mechanical deployment test after the final panel has been printed and assembled. This would help identify changes in the release path that could occur during the final manufacturing or integration process.

The launch operations plan should include a defined final configuration freeze and sufficient time for verification before the balloon is released. Components that are replaced or modified on launch day should be inspected and tested before they are incorporated into the final flight configuration. The team should also maintain a contingency procedure for external equipment failures, such as the rope-reeling mechanism, so that the launch team can respond without introducing unnecessary risk to the payload or changing the operating sequence without a documented assessment.

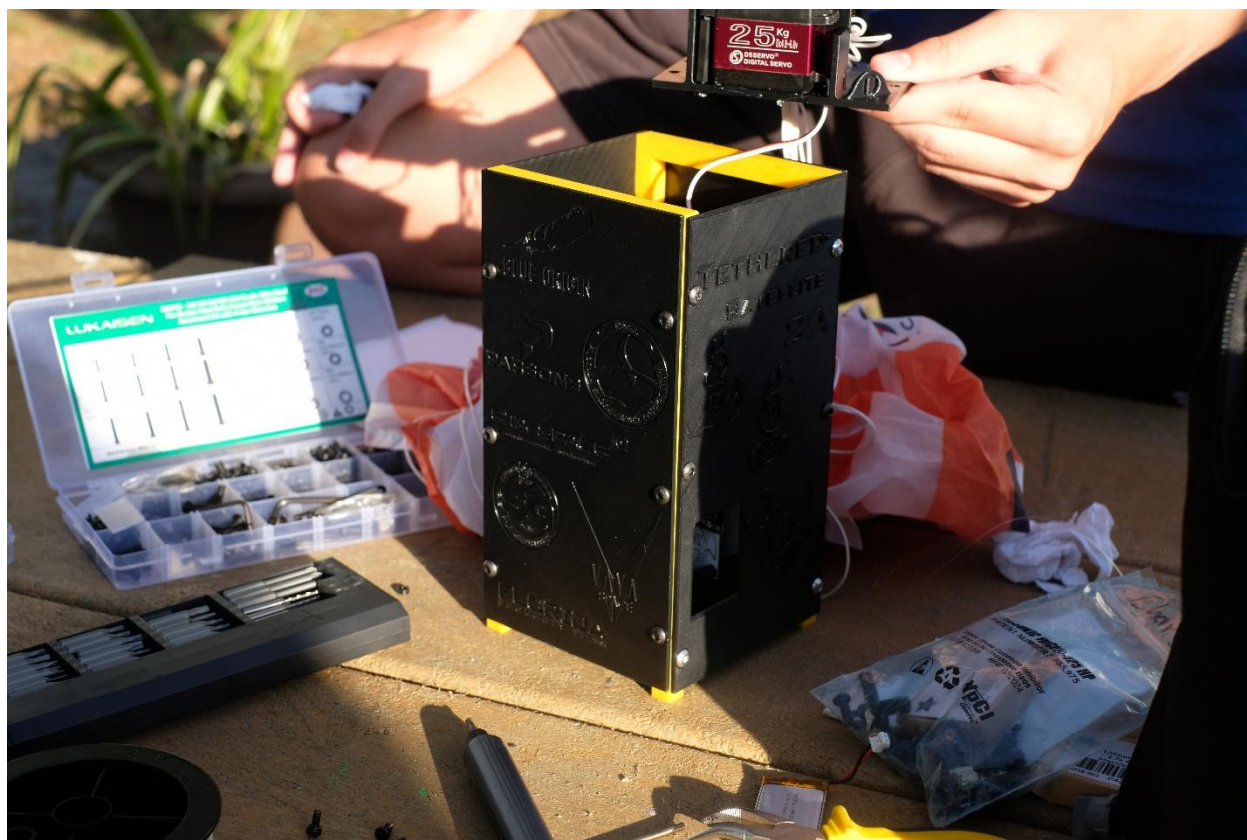
Conclusion

The Tethered Satellite 2 Mission A represented a significant engineering effort by the Knight's Satellite Club to develop, integrate, and operate a more capable CubeSat-style system. Through the final week of preparation, the team identified and addressed several mechanical, electrical, and communication concerns while preparing the vehicle for launch. The mission demonstrated successful operation of portions of the integrated flight system, including imaging, altitude data recording, and eventual physical separation from the balloon assembly. However,

the initial release attempt and subsequent parachute deployment did not perform as intended, resulting in a fall that damaged the housing.

The postflight inspection provided valuable information regarding the condition of the recovered hardware and the potential interaction between the printed structure and the parachute. The presence of recovered flight data, including images and altitude measurements, allowed the team to preserve evidence of the mission's operation despite the structural damage. These results provide a technical basis for improving future separation mechanisms, parachute interfaces, final integration procedures, and launch operations.

T-Sat 2A demonstrated the challenges of transitioning from a tested design to a complete operational flight system. The experience gained through the successful and unsuccessful aspects of the mission can support future KSC members in developing more reliable hardware, improving readiness assessments, and conducting more thorough final verification before launch. The mission's results should therefore be retained as part of the club's engineering record and used to guide future iterations of the Tethered-Satellite program.



T-Sat 2A Final Moments

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